WORKING SESSION

Building a Lead Enrichment Pipeline with LLM Agents

Relating the tactical (sales) to the technical (LLM systems), **modeled on Novara's production pipeline**

Discover > Enrich > Score > Gate > Learn

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THE PROBLEM

The job we're automating: the SDR research loop

1. **Who are the accounts?**
   Find clinics (or coffee shops) that might buy
2. **Who's the right human?**
   The operational buyer, not the figurehead
3. **Are they even a fit?**
   Qualify against an ideal-customer definition
4. **What do we say?**
   A reason THIS account should care, in their words

The thesis

We didn't build an AI that does sales.

We built a **state machine** an AI operates inside **guardrails**, with a **human holding the pen** on the one irreversible step.

SHARED LANGUAGE

Sales vocabulary, mapped to the system

Term	Sales meaning	In the system
Lead	A shop that MIGHT buy	A row in state 'To Research'
Prospect	A lead worth pursuing	A scored lead past the threshold
ICP	Who we sell to	A weighted scoring rubric
Champion	Feels the pain, advocates	Entry-point role label
Gatekeeper	Screens access	Route-around role label
Enrichment	Filling in missing facts	The model + tools job
Dedup	Don't re-add what we have	A correctness gate
Cadence	The scripted outreach sequence	Downstream of this pipeline

THE ARC

Five questions, answered twice (sales, then systems)

1	What is a lead, and what states can it be in?	State machine + memory
2	How do we fill in what we don't know?	Enrichment
3	What does the agent actually call?	Tools
4	How do we stop it doing something dumb?	Gates
5	Where does the human sit?	Human-in-the-loop + learning

1 · CORE MEMORY

The system's memory lives outside the model

STATE memory → databases

Pipeline stage, contacts, scores, signals.
Authoritative, changes daily. **Queried fresh every run, never 'remembered.'**

NARRATIVE memory → markdown

Positioning, what worked, methodology.
Durable, slow-changing. **Injected on demand for context.**

The state lives in 5 real databases:

Clinic Outreach (accounts)	Sales Contacts (humans)	Market Intel (signals)	Cron Runs (learning log)	Cases (work queue)

Status is the gate: a clinic in 'In Discussion' is off-limits to the cold pipeline. The schema's enums constrain what the model can write.

1 · CORE MEMORY

Definitions = injectable context modules

```
definitions/
    lead.md  what a lead IS
    prospect.md  the promotion rule
    icp.md  ICP as a rubric
    buyer-roles.md  champion / gatekeeper...
    state-machine.md  states + transitions
    glossary.md  shared vocabulary
```

Don't write one giant system prompt.

1. **Inject only what the worker needs**

   The scoring worker gets icp.md; the contact worker gets buyer-roles.md. Neither carries the other's baggage.

2. **Editable, versioned units**

   Learn 'we mis-score chains' → edit ONE file. The eval loop's governance edits this same surface.

3. **The ICP file is a rubric, not prose**

   Its format enforces the determinism boundary: the model retrieves facts, code computes the score.

LLM best practice: progressive disclosure / just-in-time context.

2 · ENRICHMENT

An orchestrator that runs focused workers, in waves

Wave 0	**Eval synthesis**	Read last 10 runs, self-tune
Wave 1	**Discovery (parallel)**	3 blind workers: registry · signals · intel
Gate	**Merge · dedup · score**	Cap at 10. Code, not model.
Wave 2	**Enrichment (batched)**	Clinic data + contacts + emails
Gate	**Validate · gen Angle**	Min: site + location + 1 contact
Wave 3	**HUMAN GATE (blocking)**	Ranked dashboard → approvals
Wave 4	**Write + log + self-eval**	CRM write, then grade the run

The whole game: what's code vs what's the model

CODE owns (correctness)

- Lead scoring (weighted rubric)
- Dedup (ID / domain / fuzzy name)
- Schema validation + enums
- Thresholds → priority → action

MODEL owns (judgment)

- Gathering & grounding facts
- Classifying roles from titles
- Writing the account-specific Angle
- Parsing human approval intent

The Angle: generative, but constrained by a good/bad exemplar

GOOD "280 cycles/yr across 4 REIs ≈ 70 per doctor, coordinator bottleneck. Novara automates between-visit engagement."

BAD *"Fertility clinic that could benefit from patient engagement software."*

3 · TOOLS

What the agent actually calls to touch reality

WebSearch / Fetch

Registries, job boards, news, LinkedIn profiles

Hunter.io

Find & verify emails, returns a confidence score

PostHog

Are they already on our site? Inbound = warmer lead

Notion (MCP + REST)

Read/write state; MCP for ergonomics, REST for the long tail

Subagent tool

The orchestrator's 'fork', spawn focused workers

Filesystem

Inject the narrative/definition markdown on demand

Two lessons: MCP is the 'USB-C' of tools · carry confidence/provenance as a first-class field, never collapse it.

4 · GATES

Four checkpoints, match strictness to blast radius

Schema validation Pre-flight	Do the CRM fields exist, by exact name?	**HALT before any work**
Dedup gate After discovery	Already in the CRM? (ID / domain / name)	**Exact → drop; fuzzy → ask**
Validation gate After enrichment	Min viable data present?	**Demote + flag, don't drop silently**
Human gate Wave 3	Does a human approve the writes?	**No approval → no writes**

Automate the confident & reversible. Gate the uncertain & irreversible. Fail loud, never silent.

Two human roles, keep them separate

Human as DECISION-MAKER

The approval gate (Wave 3)

Pipeline does the expensive work, then STOPS and presents a ranked dashboard.

Interface is a conversation, not a form:

```
"approve all"
"approve 1,3,5"
"edit 1: mention their EMR migration"
"not new: 2 is a dupe"
```

Human as TEACHER

The eval loop

AI grades itself first (so the human just reacts), then a 5-question eval with self-assessment pre-filled.

Accepts terse input, **"good"** or **"mostly noise"** maps to structured fields, logged to Cron Runs.

That log feeds the NEXT run's Wave 0. The loop closes.

Self-tuning under graduated governance

TIER 1 Autonomous nudges	Max 3 per run, per-run only. 'Southwest 0% for 3 runs → focus there.' Written to temp, injected into prompts.
TIER 2 Escalate to human	Persistent patterns it can't fix alone. 'Angles generic 3 runs straight → change the instructions. Approve / Dismiss / Defer?'
TIER 3 Commit the change	Approved Tier-2 edits are written to the skill file with a governance log. The system edits its own rules, with a human approving the diff.

LLM best practice: closed-loop evals + graduated autonomy. Improve without drifting.

REAL ARTIFACT

How a durable record looks in production

`pm-knowledge/prospects/terra-fertility.md  (injected when this deal comes up)`

ICP fit, as the file states it

"Terra is the textbook Novara ICP"

- Independent + physician-owner-led
- Tech-forward by stated positioning
- Founded after a PE event
- Scaling patient experience w/o staff burden

Buyer constraint that became the Angle

"No second surface."

Pietro: 'I don't want clinical users checking multiple applications.'

Decision-maker: **Dr. Pietro Bortoletto (founder)**

Real, grounded, role-tagged, not a template.

END-TO-END

One coffee shop, four stages (BrewOps demo kit)

01	**Raw discovery**	Name + city + 3 signals. Found by 2 workers (multi-source). contacts: []
02	**Enriched + scored**	540 tx/day, Square-only. Dana = Champion (verified email). Code computes 96/100.
03	**Human gate**	Ranked dashboard → STOPS. Human: 'approve 1, lead the angle with the 2nd location.'
04	**Write + learn**	Record written w/ the edit, every fact sourced. Run grades itself → logged.

Every stage adds information under a constraint; the one irreversible action waits for a human.

TAKE IT HOME

Build your own: three friction paths

A

Copilot

Claude.ai Project · Custom GPT

You are the runtime

- Paste your definitions as context
- Web-search inside the chat
- You do the CRM writes by hand

~5 min · no keys

B

START HERE

Local Agent

Claude Code · Cowork

Your laptop runs it, you approve

- Real tool calls: web · Notion · Hunter
- Stops at the human gate
- Mirrors how Novara runs it

~30–45 min

C

Always-On

OpenClaw · Hermes (self-hosted)

Runs itself on a schedule

- Cron + memory + MCP, batteries-in
- Approve from Slack / your phone
- What Novara actually runs

~half a day

Friction up, leverage up. Pick the lowest tier that does the job; graduate when the loop annoys you. → brewops/setup/

THE WHOLE THING

Tactical → Technical → LLM best practice

Sales (tactical)	System (technical)	LLM best practice
Lead lifecycle / stages	State machine over a typed CRM	External memory tames non-determinism
Word definitions	Per-concept files injected on demand	Progressive disclosure
SDR research	Orchestrator + parallel workers	Supervisor pattern; context isolation
ICP / lead scoring	Deterministic rubric + thresholds	Correctness in code, judgment in model
The pitch / Angle	Constrained generation, exemplars	Few-shot, grounded in retrieval
Don't add a dupe / junk	Schema · dedup · validation · human	Guardrails; calibrated autonomy
Manager reviews the list	Wave 3 blocking approval	HITL at the irreversible step
Coaching the rep	Self-eval + 3-tier self-tuning	Closed-loop evals; governance

THE TAKEAWAY

The model is ~20% of this system.
The state machine, the gates, and the eval loop are the load-bearing 80%.

Don't build an AI that does the job. Build the rails the AI runs on, and own the gates.

Discover > Enrich > Score > Gate > Learn